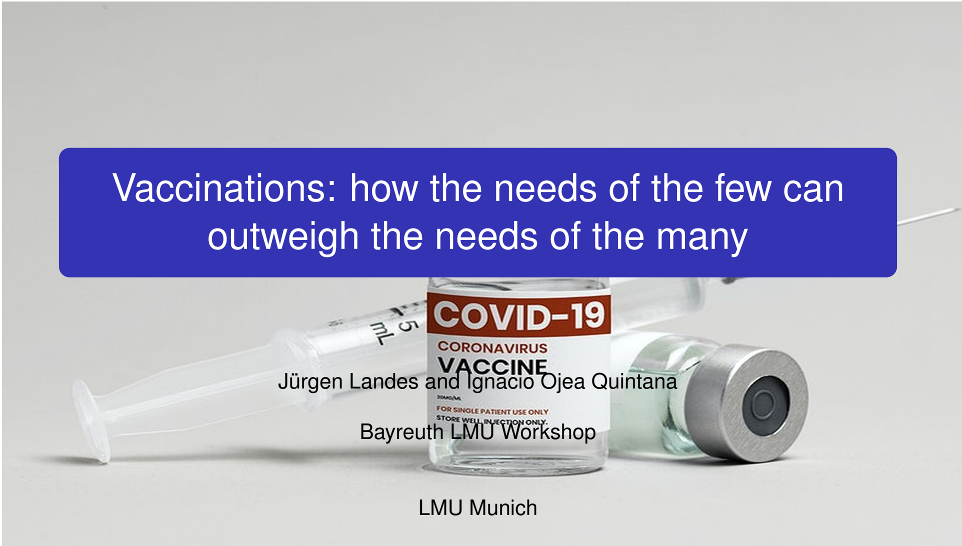

Vaccinations: how the needs of the few can outweigh the needs of the many

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Bayreuth LMU Workshop

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Plan for Today

Outline

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- Refusal to get vaccinated was deemed ethically blameworthy (unfair free-riding). [2]
- So, vaxxing the entire population is always good; right?

The Role of Models

Transmission models were used to devise vaccination strategies and served as a basis for policymaking in many countries[33,41–43]. The model-based assessment of the vaccination impact incorporates biological characteristics of the virus, age-dependent effects in transmission and disease severity, epidemiological effects of vaccines, and additional factors that are setting-specific (e.g., the level of non-pharmaceutical interventions, vaccine supply, and SARS-CoV-2 seroprevalence prior to vaccination rollout[29]). [3, P. 3]

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Virus Mutations

Prior respiratory pandemics showed development in waves and the pandemic phase extended over a 2-3 years course. The waves were characterized by different clinical severity and the pandemic phases seem to end with an attenuation of the viral virulence. [..] Even without efficient vaccines and drugs, the pandemic phase of the Russian and Spanish flu ended after a wave associated with milder symptoms. [4, P. 1314]

Transmission-virulence trade-off hypothesis

In plain words, if a pathogen causes its host to lay in bed or even to die instead of happily going around and transmitting the infection, it will be outcompeted by mutants that cause less damage to the host and allows it to produce more pathogens and infect other hosts. [5, P. 71]

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- We compare different vaccination strategies:
  1. Vax all.
  2. Vax only vulnerable.

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- Should we (*central planner*) vaccinate all?

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    - that agents are initially susceptible (S) to infection (I),
    - and an infection can have two mutually exclusive outcomes: either the agent eventually recovers and becomes susceptible again (S) or the agent dies (D).

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- Whenever an infected agent and a susceptible agent occupy the same location, then the susceptible agent is infected with some probability.
- At each time step infected agents either die with some probability or have an increasing probability of recovery.

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- **Viral age** of a virus increases by one every time a virus infects (jumps) from an infected agent to a susceptible agent.
- **Lethality** decreases with viral age (transmission-virulence trade-off), while transmission probability and recovery time remain constant.

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- Infection probability: 0.25. This is the base probability (no immunity, no vaccination, viral age zero, non-vulnerable) of virus transmission of an infected agent and a susceptible agent occupy the same grid location.

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- Death probability: 0.01. This is the base probability (no immunity, no vaccination, viral age zero, non-vulnerable) of an infected agent dying per daily time step.
- Recovery time: 30. Agents keep track of how long they have been infected, and as time progresses they have increased probability of recovery, reaching probability 1 at recovery time.

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- Viral ageing: 0.1, every time the virus is transferred to an agent, its death probability is reduced by 10% (but infectivity and recovery time are uneffected).

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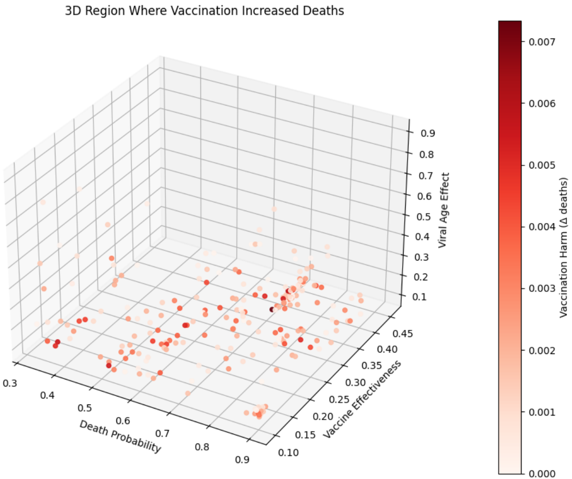
2. *Vaccination effect* in [0.1,0.9].
3. *Death probability* in [0.01,0.9].

Parameter Settings where it is better not to vaccinate all



Context for Figure

- The reason to use trials ($n = 200$) is to avoid the possibility that Strategy 2 was outperforming Strategy 1 due to fortuitous chance events, since our model uses a number of random choices (initial distribution of agents in the grid, random walks, transmission probabilities, etc.).

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- We employed the *scikit-optimize* package using Bayesian Optimisation and a Gaussian Process (GP) regression model to minimise our objective function, over a total of $k = 1000$ parameter settings.
- Minimising the objective function we found 224 triples of parameters (*Viral ageing*, *Vaccination effect* and *Death probability*), for which the objective function was negative: Strategy 2 is preferable to Strategy 1.

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Difference between Strategies

1. The average difference between the strategies is never large (the maximal difference in proportions of dead agents was 0.7%).
2. While there are some areas of the parameter space in which none of the tested parameters lie, the tested parameters are spread widely across multiple regions.
3. We also ran simulations which vary more parameters and further points were found, but these are hard to visualise, and there did not seem to be any systematic relation between the parameter variables for where Strategy 2 was better than Strategy 1.

On Predictability

1. It is not easy to predict when vaccinating all is the better strategy.

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1. It is not easy to predict when vaccinating all is the better strategy.
2. It is hard to be sure that Strategy 2 is worse in a concrete situation.

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- Previous criticisms were often made on principled grounds and/or on concrete case studies.
- We instead identify a biological mechanism that is (mostly) missing in theories and models, which we integrate into our own model.
- Since our model seems no less reasonable than previous models and the results from our model conflict with the received wisdom, we argue that we should be very careful when drawing conclusions from pandemic models.

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- To achieve the an outcome preferable to the community, a majority has to accept a greater risk to their well-being.
- “The needs of the few outweigh the needs of the many.”

Future Work

- Compare deaths in vulnerable and non-vulnerable sub-groups.

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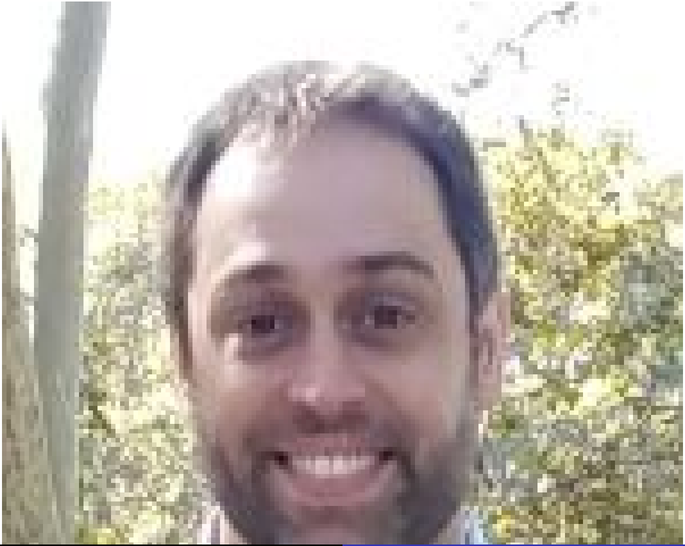
- Make the model more complex – apparently....

Thank You!



P. Singer, Why Vaccination Should be Compulsory, Project Syndicate (2021).
URL https://www.project-syndicate.org/commentary/why-covid-vaccine-should-be-compulsory-by-peter-

URL
https://doi.org/10.1038/s43856-023-00320-x

H. Brüssow, The beginning and ending of a respiratory viral pandemic-lessons from the Spanish flu, Microbial Biotechnology 15 (5) (2022) 1301–1317.
URL https://doi.org/10.1111/1751-7915.14053

A. Kun, A. G. Hubai, A. Král, J. Mokos, B. A. Mikulecz, A. Radványi, Do pathogens always evolve to be less virulent? The virulence-transmission trade-off in light of the COVID-19 pandemic, Biologia Futura 74 (1-2) (2023) 69–80.
URL
https://doi.org/10.1007/s42977-023-00159-2